EA-Sindy notes

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Setup: Let $\dot{x} = f(x)$ be some complex high dimenstional system with support in $\mathcal{X}$, and $X, \dot{X} \in \mathbb{R}^{N \times D}$ be data sampled from said system. To begin we train an ensemble of boostrap models on this data. We sample indexes $\{1, ..N\}$ with replacement to create b data bags of size M. For $i = 1, ..b$, bag index i is associated with indexes $\{i_1, ...i_M\} \in \{1, ...N\}^M$, yielding data bags, $X^{(i)}, \dot{X}^{(i)} \in \mathbb{R}^{N \times M}$. We train sindy models on each of bags $(X^{(i)}, \dot{X}^{(i)})$ yielding model params:
I. Encoder NN $E^{(i)} : \mathbb{R}^D \to \mathcal{Z}^{(i)} \subset \mathbb{R}^d$, II. Decoder NN $D^{(i)} : \mathcal{Z}^{(i)} \to \mathbb{R}^D$
III. Sindy coefficients $\Xi^{(i)} \in \mathbb{R}^{d \times p}$, IV. Coefficient mask $\Phi^{(i)} \in \{0, 1\}^{d \times p}$
Our objective is to obtain a single model
$\tilde{\mathcal{S}} = (\tilde{E}, \tilde{D}, \tilde{\Xi}, \Phi)$ which combines models $\mathcal{S}^{(i)} = (E^{(i)}, D^{(i)}, \Xi^{(i)}, \Phi^{(i)})$, for $i = 1, ...b$ and yields lower test errors. Naive combination of models $\{\mathcal{S}^{(i)}\}_{i=1}^b$ via model output averaging may work in achieving lower test error but it would sacrifice the interpretability of having a single latent space $\mathcal{Z}$ and a single description of lantent dynamics Ξ, compromising the whole interpretable coordinate discovery purpose of auto-encoder sindy.

Instead what we'll attempt to do is to find a way to 'translate' the latent spaces $\{\mathcal{Z}^{(i)}\}_{i=1}^b$ into a single latent space $\mathcal{Z}$, in which we can apply standard linear ensemble regularization techniques to combine the dynamics learnt by the various sindy models.

Our strategy to achieve this will be to look for NN maps
$F^{(i)} : \mathcal{Z}^{(i)} \to \mathcal{Z}$ ('translators') $D : \mathcal{Z} \to R^D$, and sindy coefficients $\Xi^{(i)}$ s.t for any i:
i. Network $F^{(i)}$ translates latent space $\mathcal{Z}^{(i)}$ to a shared latent space $\mathcal{Z}$.
ii. Network D decodes latent space $\mathcal{Z}$ back to $\mathbb{R}^D$
iii. $\Xi^{(i)}$ describes the time dynamics of $z = F^{(i)}(E^{(i)}(x))$
iv. $\Xi^{(i)}$ implicitely describes the time dynamics of $\tilde{x} = D((F^{(i)}(E^{(i)}(x))))$
v. $\Xi^{(i)}$ is sparse in the L_0 sense.
In other words, $\mathcal{S}^{(i)} = (F^{(i)} \circ E^{(i)}, D, \Xi^{(i)})$ is a valid sindy model.

For $i = 1, ...b$ we denote $\tilde{E}^{(i)} = F^{(i)} \circ E^{(i)}$. We may enforce criteria i-v by minimizing loss :

$$\tilde{\mathcal{L}}(\{F^{(i)}, \Xi^{(i)}\}_{i=1}^{b}, D; X, \dot{X}, \{E^{(i)}\}_{i=1}^{b}) = \mathbb{E}_i[\mathcal{L}_{\text{sindy}}(\tilde{E}^{(i)}, \Xi^{(i),D}; X, \dot{X})] + \lambda \mathbb{E}_{i,j}[(\text{corr}(\tilde{E}^{(i)}(X), \tilde{E}^{(j)}(X))]$$

Minimizing $\tilde{\mathcal{L}}$ yields translator networks $F^{(i)}$, and decoder D s.t for any index $i = 1, ...b$ $\mathcal{S}^{(i,} = (F^{(i)} \circ E^{(i)}, D, \Xi^{(i)})$ is a valid sindy model. The only way for all these models to be cross compatible with eachother is for all of them to map into the same shared latent space $\mathcal{Z}$ and for each of sparse coefficients matrixes $\Xi^{(i)}$ to describe the dynamics in $\mathcal{Z}$. In order to get this to work in pratice we also need to add a term which explicitly incentives the latent code for a given input x derivered by models i and j to be highly correlated. Hence why we include the correlation term $\mathbb{E}_{i,j}[(\text{corr}(\tilde{E}^{(i)}(X), \tilde{E}^{(j)}(X))]$. Without this term the model error sometimes blows up, I'm not yet sure why, but there is precedent for the neccesity of this kind of regularization in latent space combination schemes.
Note, this frameworks corresponds very closely to correlation base methods in mutli-view learning such as DCCA correlation neural networks, just treating the different learnt representations $z^{(i)}$ of an input x as different 'views' of some underlying intrinsic code $\tilde{z}$. Also note, the optimization of objective $\tilde{\mathcal{L}}$ is done only, w.r.t the parameters of $F^{(i)}, D, \Xi^{(i)}$. The parameters of $E^{(i)}$ are regarded as constants. Given a set of inter-compatible models $\{\mathcal{S}^{(i)} = (\tilde{E}^{(i)}, D, \Xi^{(i)})\}_{i=1}^{b}$, all associated to the same latent space $\mathcal{Z}$, me way apply naive model averaging in $\mathcal{Z}$ to obtain ensemble model parameters

$$\text{I. } \tilde{E} := \mathbb{E}_i[\tilde{E}^{(i)}], \text{ II. } \tilde{\Xi} = \mathbb{E}_i[\tilde{\Xi}^{(i)}] \tag{1}$$

In practice, we find the standard lasso trick of penalizing the 1-norm of the cofficient matrixes Ξ does not suffice for finding L-0 sparse solutions. The usual sindy trick is to apply sequentially thresholded least squares, whereby we maintain some coefficient mask $\Phi \in \{0, 1\}^{d \times P}$ and update it iteratively as to loss minimization proceeds, as to 0 out sindy coefficents close to 0. Our ultimate goal is for the ensemble coefficient matrix $\tilde{\Xi} = \mathbb{E}_i[\tilde{\Xi}^{(i)}]$ to be L-0 sparse. We may view the ensemble coefficient matrixes $\{\tilde{\Xi}^{(i)}\}_{i=1}^{b}$ as b different descriptions of the same dynamics in $\mathcal{Z}$. Having this sample of possible coefficients allows us to use stability selection as the coefficient selection criterion in our sequentially thresholded least squares procedure . In other words, instead of updating the ensemble mask $\tilde{\Phi}$ via 'averaging' thresholding update rule given

$$\tilde{\Phi}_{k,j} \leftarrow \mathbb{1}\{|\tilde{\Xi_{k,j}}| > \varepsilon_0\} * \tilde{\Phi}_{k,j}, \text{ for } \varepsilon_0 > 0 \tag{2}$$

We instead us the the stability selection update rule given

$$\tilde{\Phi}_{k,j} \leftarrow \mathbb{1}\{\mathbb{E}_i[\mathbb{1}\{|\Xi_{k,j}^{(i)}| > \varepsilon_0\}] > \varepsilon_1\} * \tilde{\Phi}_{k,j}, \text{ for } \varepsilon_0 > 0, \text{ and } \varepsilon_1 \in (0, 1) \tag{3}$$

There is evidence in the literature that this is a superior coefficient selection criterion obtaining sparse solutions. The full architecture is given by fig 1.

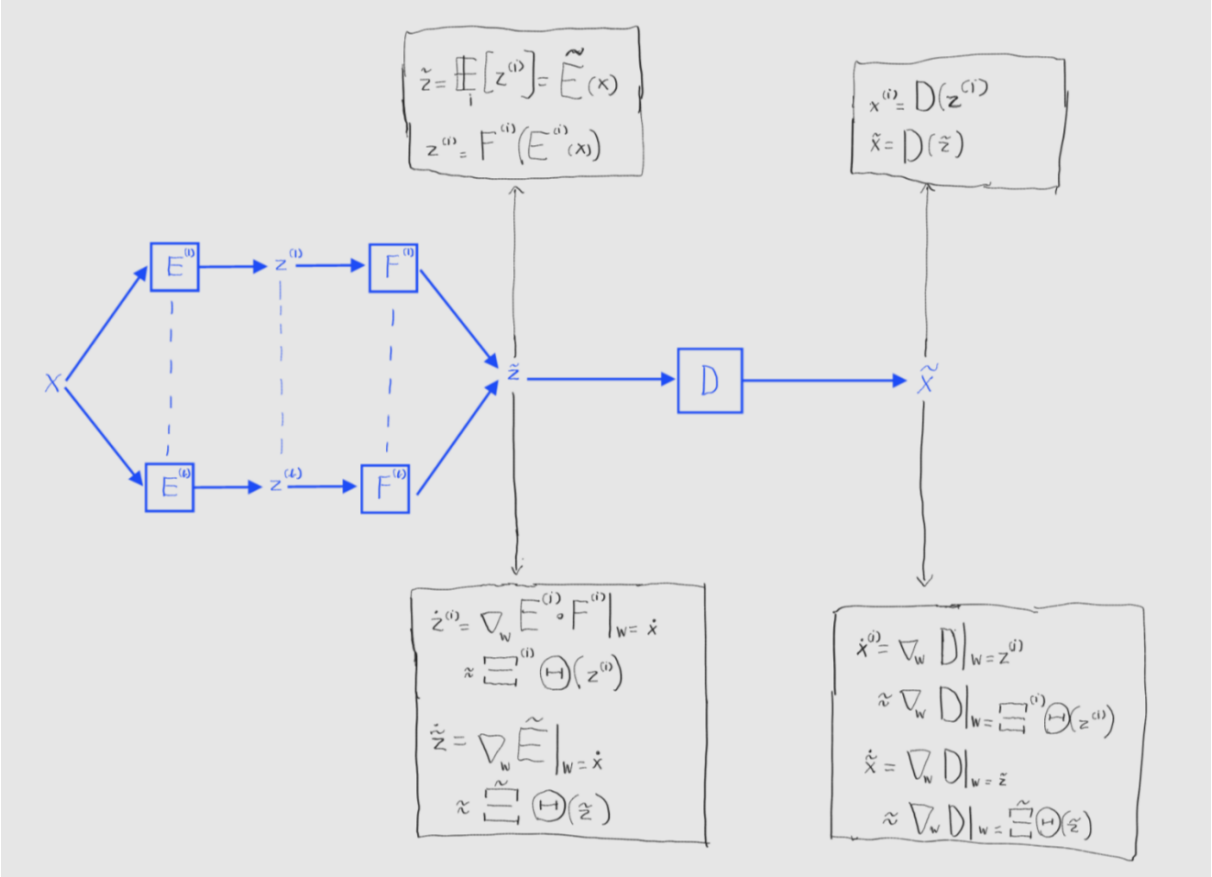


Figure 1: Model achitecture schematic for latent ensemble model

Numerical results:

We sample $(X, \dot{X}) \in \mathbb{R}^{50000\times 128}$ from lorentz system example from sindy autoencoders paper. We assume latent cordinate space of dimension $d = 3$.

For $i = 1, ...20$ we train sindy model $((E^{(i)}, D^{(i)}, \Xi^{(i)}, \Phi^{(i)})$ to convergence on data bag $(X^{(i)}, \dot{X}^{(i)}) \in \mathbb{R}^{50000\times 128}$ drawn with replacement from $(X, \dot{X})$.

We define the naive ensemble model to be given as the average output of models i = 1,...10.

More precisely we have the naive ensemble loss are given:

i. $\mathcal{L}^{(n)}_{\text{decode}}(x) = \|x - \mathbb{E}_i[(D^{(i)}(E^{(i)}(x)))]\|_2^2$ ii. $\mathcal{L}^{(n)}_{dx}(x) = \|\dot{x} - \mathbb{E}_i[\nabla_z D^{(i)}((\Phi^{(i)} \odot \Xi^{(i)})(E^{(i)}(x)))]\|_2^2$

We call this ensemble naive because, unlike the latent ensemble, it does not combine latent spaces $Z^{(1)}, ...Z^{(10)}$ into a collective latent space Z.

To create the latent ensemble we randomly initialize ensemble components $F^{(i)}, D, \Xi^{(i)}$, for $i, = 1, ...10$. Each of $F^{(i)} : \mathbb{R}^3 \to \mathbb{R}^3$ is paramatrized with Resnet. We update the paramaters of $(F^{(i)}, D, \Xi^{(i)}), i = 1, ...10$ via SGD on loss $\tilde{\mathcal{L}}$ until convergence. Coefficient mask $\tilde{\Phi}$ is updated via stability selection update rule (4) every 500 training epochs. The latent ensemble losses are given by

iii. $\mathcal{L}^{(l)}_{\text{decode}}(x) = \|x - \tilde{E}[(\tilde{D}(x)]\|_2^2$ iv. $\mathcal{L}^{(l)}_{dx}(x) = \|\dot{x} - \nabla_z D((\tilde{\Phi} \odot \tilde{\Xi})(\tilde{E}(x)))\|_2^2$

Where the ensemble encoder $\tilde{E}$ and coefficient matrix $\tilde{\Xi}$ are given by (2)

We plot the trajectories for each of these losses, as well as the number of active coefficients per epoch for the base sindy models and latent ensembles.

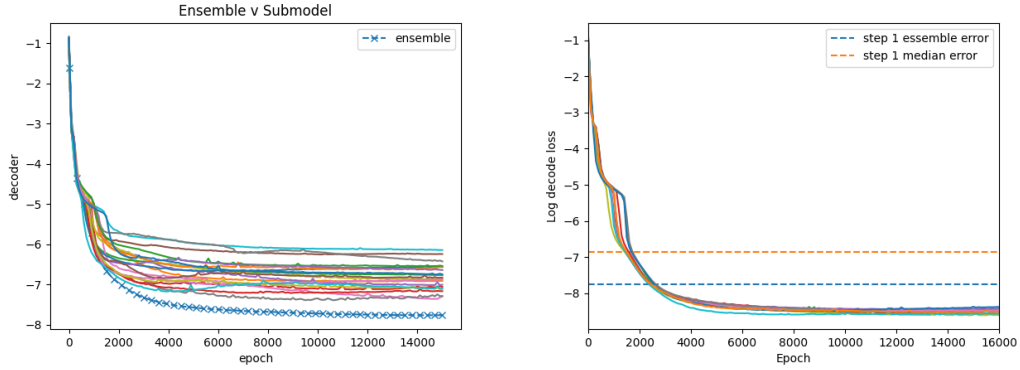


Figure 2: (Left) Decoder loss trajectories for 10 Sindy models on Lorentz system, compared with naive ensemble decoder loss ($\mathcal{L}^{(n)}_{\text{decode}}$) trajectory . Ensemble loss is given by line with x markers. (Right). Latent ensemble Decoder loss ($\mathcal{L}^{(l)}_{\text{decode}}$) trajectories for 10 latent ensemble Sindy models. Each of the 10 latent ensemble models was built using the base Sindy models used to make left plot. Errors are test errors.

Figure 2 shows firstly that the latent ensemble well outperforms any of the constituent models in reconstruction error and secondly that the latent ensemble model achieved lower deconstruction error than the naive ensemble in 10/10 trials.

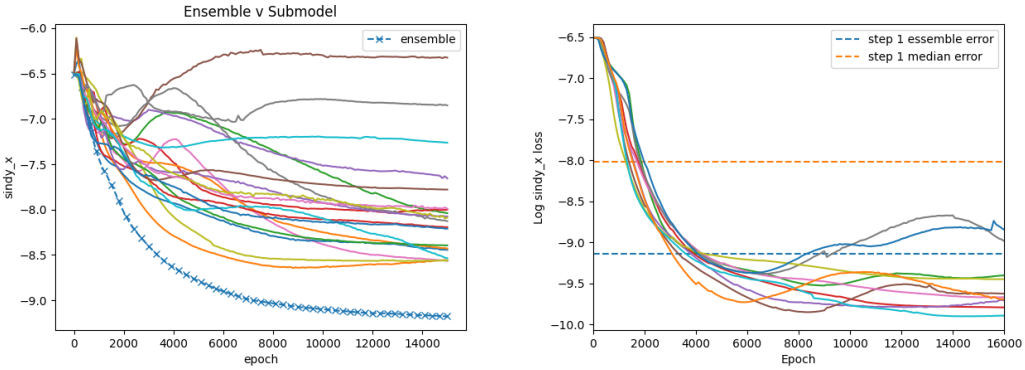


Figure 3: (Left) Sindy x loss trajectories for 10 Sindy models on lorentz system, compared with naive ensemble Sindy x loss ($\mathcal{L}^{(n)}_{\text{dx}}$) trajectory . (Right). Latent ensemble Sindy x loss ($\mathcal{L}^{(l)}_{\text{dx}}$) trajectories for 10 latent ensemble Sindy models. Each of the 10 latent ensemble models was built using the base Sindy models used to make left plot. Errors are test errors.

Figure 3 shows firstly that the latent ensemble well outperforms any of its constituent sindy models and secondly that the latent ensemble model achieved lower final test Sindy x loss than the naive ensemble 8/10 trials.

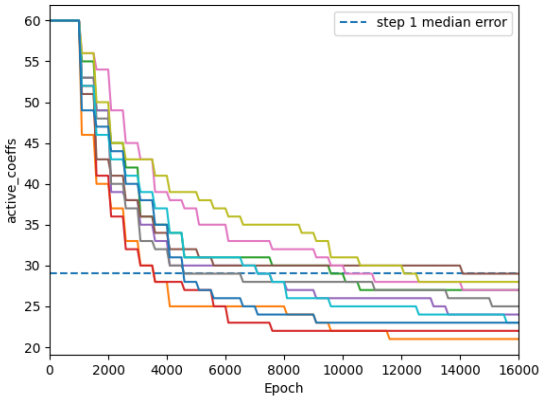


Figure 4: Active coefficients per epoch for 10 latent ensemble Sindy models. Each of the 10 latent ensemble models was built using the base Sindy models used to make left plot.

Figure 4 shows the latent ensemble model achieved superior/equivalent final sparsity than any of the 10 sub-models in all 10 trials.

The importance of these results is it shows how doing ensemble regularization in the latent space enables us to build a single encoder decoder model $\tilde{S} = (\tilde{E}, D, \tilde{\Phi})$ which achieves far superior performance(in terms of all relevant losses and sparsity) to any of its constituents and does not sacrifice interpret-ability/ information bottle necking like the naive ensemble. Optimizing the ensemble model is also far more stable/less prone to overfitting than fitting the sub models, as show by the test error curves.

I am interested in the connections between this ensemble based approach and bayesian autodencoder sindy, given the natural connection between bayesian methods and ensemble methods. Namely, I'm looking into wether we can match the performance of the bayesian model with less compute using an ensemble and what more rigorous mathematical connections we can draw between the two approaches.